CODENEXUS / NSEC / TEAM CHROLLO

Dante

Brand change and drift tracking

A source-linked review workspace for teams that need every claim to stay current.

The problem Dante solves

Product facts, brand rules, approved terms, and visual assets often live in separate files and channels. New content can repeat outdated claims, while older material stays online after the underlying fact changes. Teams need to find affected items, see evidence, and route decisions to a person.

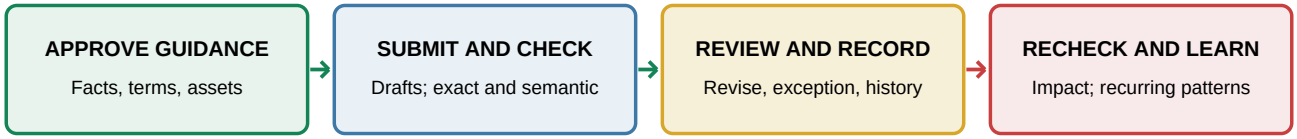
Why this problem matters

Lucidpress, *The State of Brand Consistency* (2019), surveyed 200+ organizations. Historical, self-reported results; the two questions are distinct and are not a current benchmark. Open report.

Who it serves

TEAM	NEED
Brand and marketing	Keep approved claims consistent across campaigns and channels.
Creators and designers	Receive specific evidence and a useful correction path.
Brand reviewers	Assess evidence, decide exceptions, and track changes.
Product and communications	Update claims after facts or policies change.
Distributed teams	Share current guidance with employees and agencies.

How the product works



Full sequence: add guidance, approve its version, submit text and channel, check evidence, revise or record an exception, save the decision, recheck linked content after source changes, and review recurring patterns.

Demo scenario: an approved storage claim changes from 100 GB to 200 GB. Dante identifies linked drafts that still say 100 GB, shows the source version and evidence, and sends each item to a reviewer.

Team: Samriddha Chaudhury, Team Lead and Full Stack Developer; Md Rizwaan Rahaman, Full Stack Developer. Contact: samriddhachaudhury@gmail.com.

Major features Dante offers

A working workspace for governed content review, with clear boundaries between automated checks and human decisions.

Approved source library

Create facts, rules, required terms, and logo assets. Keep approved versions and history.

Evidence-linked content review

Run exact quantity and term checks. Optional Gemini rule review returns source and draft quotes for a person to assess.

Change impact analysis

Preview linked content affected by a source update; stale or conflicting content returns to review.

Human review and exceptions

Reviewers revise content or record a reasoned exception. Source changes and decisions enter the audit trail.

PDF and DOCX policy intake

Extract likely rules with page or paragraph references. Administrators approve candidates before they become active.

Scanned PDF OCR

When a page has little selectable text, the uploader can opt in to Gemini OCR. Candidates remain unapproved and carry a verification warning.

Campaign image review

With explicit consent, Gemini transcribes visible copy and gives advisory logo and contrast findings; an approved logo image can be compared.

Public page monitor

Schedule checks of public server-readable pages every 15 minutes, hourly, or daily. Initial snapshots and changes require reviewer decisions.

Analytics and quality evaluation

Recharts show workspace activity. A confusion matrix combines curated cases with reviewer-labeled examples to report precision, recall, false alarms, and missed review flags.

Workspace access and audit

Supabase Auth, roles, email-bound invitations, reviewer assignment with an assigned-first queue, private files, and CSV/JSON audit export.

Publishing gate

A reviewer authorizes current content before Dante sends it to a configured generic HTTPS webhook.

Safety controls

Server-side authorization, source-version checks, opt-in Gemini calls, private storage, and reset protection for persistent workspaces.

Dante does not auto-approve extracted rules or publish content without a reviewer action. Gemini outputs remain advisory.

Round 3 focus areas and current status

The Round 2 proposal named five outcomes. This table distinguishes what is now built from what remains limited.

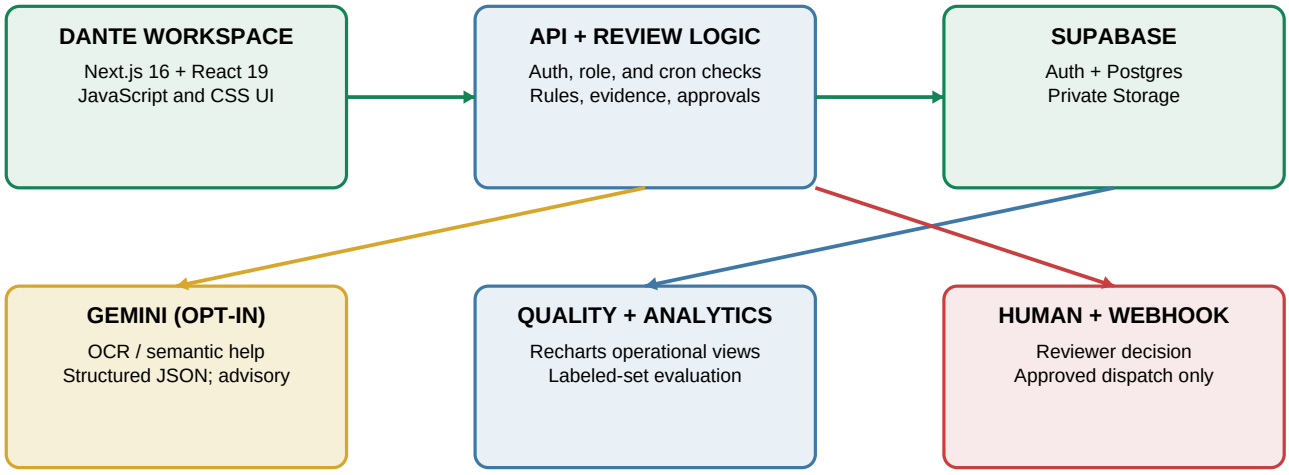
ROUND 2 COMMITMENT	STATUS	WHAT WORKS NOW / LIMITS
Connect content sources	**IMPLEMENTED**	Scheduled public-page text checks; reviewer-approved baseline and change review. Generic reviewer-gated webhook. Native CMS/OAuth connectors are still future work.
Bring in team guidance	**IMPLEMENTED**	PDF/DOCX extraction with page or paragraph references, opt-in scanned-PDF OCR, and administrator approval before activation.
Support team review	**IMPLEMENTED**	Shared workspaces, roles, reviewer assignments, and an assigned-first in-app queue. No email/push notifications or configurable multi-stage approvals.
Measure review quality	**IMPLEMENTED**	Curated fact-check baseline plus reviewer-labeled examples; confusion matrix, precision, recall, false alarms, and missed review flags. Small-set evaluation; no automatic tuning.
Broaden coverage	**IMPLEMENTED**	Multilingual rule assistance and opt-in image/logo review route uncertainty to people. Coverage is limited; visual contrast is advisory, not WCAG measured.

Round 3 work delivered

IMPROVEMENT	DELIVERED
Scheduled sources	Configured intervals, guarded Vercel cron, page-change evidence, and human approval gates.
Team workflow	Reviewer assignments, assigned-first queue, role checks, audit entries, and duplicate-action protection.
Measured quality	Reviewer labels now feed a confusion matrix and transparent precision/recall and error counts.
Guidance and coverage	Traceable PDF/DOCX/OCR intake, language-aware rule review, and consented image/logo checks.
Reliability fixes	Persistent-workspace reset is blocked; reviewer-only source monitoring and schedule endpoint guards are enforced.

Technical architecture

The browser uses the authenticated workspace API. Server routes verify the session and workspace membership before loading or saving workspace data. Optional AI calls run server-side after explicit consent.



LAYER	IMPLEMENTATION	ROLE
Application	Next.js 16.4.5, React 19, JavaScript, CSS	Authenticated shell, review UI, app routes, and workspace views.
API and rules	Next.js Route Handler plus server.mjs	Session / role checks, deterministic checks, Gemini orchestration, impact logic, and webhooks.
Data and identity	Supabase Auth and PostgreSQL	Workspace membership, sources, content, audit history, and saved state.
Files	Supabase private Storage	Original policy and logo files; admin-only short-lived document links.
AI	Gemini GenerateContent REST API	Opt-in structured OCR / semantic assistance; exact quotes are checked by the server.
Charts and deployment	Recharts; Vercel Cron	Workspace activity charts, evaluation measures, and scheduled page checks. Hosted cron and deployment still require production smoke testing.

Privacy boundary: Supabase and Gemini credentials remain server-side. The browser sends the user session to Dante; files are only forwarded to Gemini when the relevant consent is given.

Review reliability: deterministic checks handle exact facts and required terms. Gemini assists with semantic interpretation or extraction; the server validates evidence against the approved source, and unclear findings stay with a human reviewer.

Testing and reliability

CHECK	RESULT	SCOPE
Automated feature tests	**8 passed, 0 failed**	Evidence and OCR gates; private originals; multilingual quote validation; reset protection; idempotent reviewer assignment; evaluation labels and metrics; monitor role and scheduled-job guards.
Production build	**Passed**	Next.js compile, TypeScript phase, static route generation, and API route generation completed locally.

How to reproduce

```
node --test tests/round3-features.test.mjs
npm run build
```

Test limits

- Gemini is mocked in the automated tests. They verify request gating, parsing, evidence validation, and state transitions, but do not measure live-model accuracy.

- The OCR test uses a synthetic PDF with no text layer and a mocked transcript. A photographed/scanned guideline has not been evaluated against live Gemini OCR.

- The accuracy lab ships with a small curated numeric baseline. Workspace labels are supported, but no real pilot labels have been entered; these metrics are not a claim of real-world accuracy.

- Automated tests cover role gates and safe empty cron runs, not a full Vercel cron cycle, external website changes, native CMS integration, load, or production smoke testing.

- The monitor reads server-delivered HTML; JavaScript-rendered pages may be missed. Scheduled runs require CRON_SECRET and a configured Supabase service key in the hosted environment.

Current safeguards

Human approval is required for extracted rules and publishing. Stale source versions block publishing approval. Workspace routes check authentication and role; original policy files stay private; OCR and image analysis are opt-in.

Product roadmap and future development

NEXT	MEASURABLE OUTCOME
Deploy and smoke-test the final build	Verify Supabase environment keys, OCR consent, PDF upload, role checks, original-file access, and the reviewer-gated webhook in the hosted environment.
Grow and validate the accuracy set	Collect human-reviewed examples across languages and channels; report error rates with sample size, then tune only after independent validation.
Add native source connectors	Integrate a CMS with scoped permissions, scheduled sync, idempotent updates, and a review step before approved guidance changes.
Improve multilingual and visual quality	Expand verified language coverage; compare OCR and semantic findings with human judgments; add measured contrast checks where technically reliable.
Extend team governance	Add email/push notifications, configurable approval stages, retention controls, and richer audit exports.
Surface recurring issues	Group repeated findings and overrides with examples, then compare the patterns against labeled review outcomes.

Expected impact

Dante is designed to reduce time spent locating current guidance, surface stale claims before publishing, and give reviewers a traceable basis for decisions. Those benefits remain goals until measured with real teams and a labeled evaluation set.

Team

MEMBER	ROLE	CONTRIBUTION
Samriddha Chaudhury	Team Lead; Full Stack Developer	Product scope, backend, data model, source links, change-impact logic, review workflow, dashboards, and integration.
Md Rizwaan Rahaman	Tester	Demo content and testing.

Team Chrollo | CodeNexus | NSEC | samriddhachaudhury@gmail.com

Dante keeps the decision with the people who own the brand.